UIET, CSJMU KANPUR

INDUSTRIAL INTERNSHIP REPORT

Embedded Systems and IoT Training

Embedded Systems and Internet of Things Internship

Prepared by	Gayatri
Institution	UIET, Chhatrapati Shahu Ji Maharaj University (CSJMU), Kanpur
Internship	Embedded Systems and Internet of Things Internship
Project / Status	Training and assignments only; no separate project/problem statement assigned.
Repository	https://github.com/gayatri-535/upskillcampus

Prepared as internship learning and project documentation.

Prepared by Gayatri | 2026

Executive Summary

This report documents Gayatri's Embedded Systems and Internet of Things (IoT) internship learning. The learner confirmed that the work consisted of training and assignments and that no separate project/problem statement was assigned. This report documents the training rather than claiming a completed IoT product.

Topics shared include IoT deployment challenges, basic electronics and circuit design, sensors and actuators, analog-to-digital and digital-to-analog conversion, input/output devices, and circuit design terms. The learner also completed IoT Test 2.

The supplied syllabus includes embedded system architecture, ARM processor architecture, IoT sensors, platform design methodology, Raspberry Pi, Web of Things, Cloud of Things, IoT cloud offerings, and case studies such as weather monitoring, air-pollution monitoring, smart irrigation, and home intrusion detection.

Diagrams and test plans are conceptual learning aids, not evidence of a built or deployed system.

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Format adapted from the supplied report template and adjusted for training-only work.

1. Preface

This report summarizes my Embedded Systems and IoT internship learning. The training explored how electronic components, sensors, controllers, communication networks, and cloud services can work together in connected systems.

Activities included basic electronics, circuit diagrams, electrical measurements, electronic components, input/output, sensors and actuators, signal conversion, circuit design concepts, IoT deployment challenges, and IoT Test 2.

No separate project/problem statement was assigned. This report records training and potential future applications rather than claiming that a hardware or cloud-connected system was built.

Message to peers: learn physical and electrical fundamentals alongside IoT architecture because system reliability depends on both.

2. Introduction

Embedded systems are computing systems designed for specific tasks within a larger device. IoT connects physical devices so that information can be collected, exchanged, analyzed, or used to trigger actions.

The training covered electronics, sensors and actuators, input/output behavior, analog and digital signals, circuit design, connectivity, and cloud concepts.

The supplied syllabus names embedded system architecture, ARM processor architecture, IoT sensors, platform design methodology, Raspberry Pi, Web of Things, Cloud of Things, cloud offerings, and IoT case studies.

The confirmed status is training and assignments only. Case studies are learning examples, not completed deployments.

2.1 Internship Context

The supplied sample report describes an internship facilitated by UpSkill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). It introduces UCT's work in digital transformation, IoT, cloud computing, communication technologies, and industrial monitoring.

The sample mentions dashboards, alerts, analytics, smart factory monitoring, LoRaWAN-based solutions, and predictive maintenance. These are contextual examples from the sample, not projects implemented by Gayatri.

2.2 Training Provider and Learning Environment

The supplied template describes UpSkill Campus as a career-development platform and The IoT Academy as an education and training division associated with UCT. This report follows the template's general structure.

The syllabus combines foundational electronics with higher-level IoT architecture. A connected device must measure or control the physical world reliably before its data can be useful to a network or cloud application.

https://github.com/gayatri-535/upskillcampus

2.3 Objectives

Objectives of the training:

• Understand embedded systems and the role of hardware and software.

• Learn electrical quantities, circuit diagrams, and common components.

• Understand sensors, actuators, I/O devices, ADC, and DAC.

• Study circuit-design topics such as impedance, noise margin, pull-up/pull-down resistors, filters, loading effect, fan-in, and fan-out.

• Understand IoT architecture, device connectivity, cloud concepts, and deployment challenges.

• Relate concepts to case studies and complete knowledge assessments.

2.4 References and Learning Materials

Report basis: the syllabus images supplied by the learner, weekly notes on IoT deployment challenges, circuit-design training topics, IoT Test 2 activity, and the supplied internship report template.

The syllabus lists *Designing the Internet of Things* by Adrian McEwen and Hakim Cassimally; *Internet of Things (A Hands-on Approach)* by Vijay Madisetti and Arshdeep Bahga; and *Architecting the Internet of Things* by Dieter Uckelmann, Mark Harrison, and Florian Michahelles.

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2.5 Glossary

Term	Meaning
Embedded system	Computing system designed for a specific function
IoT	Connected physical devices that exchange or act on data
Sensor	Component that detects a physical quantity
Actuator	Component that performs an action in response to a control signal
ADC	Converts analog signal to digital representation
DAC	Converts digital values into an analog signal
Noise margin	Range that helps tolerate signal noise while preserving logic levels
Pull-up/down resistor	Holds a signal at a defined logic level when needed
Gateway	Connects devices or networks and may translate protocols
Cloud platform	Remote service for storing, processing, or visualizing data

3. Training Problem / Learning Need

The central learning challenge in embedded IoT is understanding how physical signals become usable information and how information moves through a complete system. Hardware, signal conditioning, processing, connectivity, and application behavior must work together.

The training progressed from basic electronics and measurement to input/output devices, circuit-design constraints, IoT architecture, cloud concepts, and case studies.

Key questions included selecting suitable sensors, understanding analog/digital signals, recognizing causes of unreliable circuits, understanding device connectivity, and considering power, environment, cost, and maintenance.

No individual project statement was provided; these points summarize the learning need addressed by training.

4. Existing Knowledge and Proposed Learning Approach

The training organizes subjects into a system-level learning path: electronics first, then input/output and signal conversion, then circuit-design constraints, and finally IoT connectivity and cloud use cases.

Learning area	Key topics	Importance
Electronics	Voltage, current, resistance, components	Supports circuit reasoning
I/O and signals	Sensors, actuators, ADC/DAC, discrete I/O	Connects physical conditions to logic
Circuit design	Impedance, noise, filters, loading, pull resistors	Improves signal reliability
IoT architecture	Devices, gateways, connectivity, cloud	Explains end-to-end data flow

This approach connects individual concepts into a complete system view without overstating practical implementation.

4.1 Training Repository Link

Repository supplied by the learner:

https://github.com/gayatri-535/upskillcampus

Since no separate IoT project was assigned, the repository can contain shareable training notes, assignments, learner-created diagrams, and this report if required.

Verify evaluator access, use clear filenames, upload only permitted materials, and do not describe training examples as a completed prototype.

4.2 Report Submission Link

Report filename: EmbeddedIoT_Gayatri_USC_UCT.pdf

Expected URL after uploading this PDF to the repository's main branch:

https://github.com/gayatri-535/upskillcampus/blob/main/EmbeddedIoT_Gayatri_USC_UCT.pdf

The URL will work only after the PDF is uploaded at that path. If the branch or folder differs, update the link and regenerate the report.

5. Proposed Conceptual IoT Model

No physical or software project was assigned. This model explains the architecture discussed in training and is not a diagram of a system built by the learner.

Stage	Role
Physical environment	Condition such as temperature, motion, moisture, or air quality
Sensor / input	Measures the condition and produces an electrical signal
Embedded controller	Reads input and applies programmed logic
Connectivity / gateway	Transfers data to another device, network, or service if needed
Cloud / application	May store, display, or analyze data and send alerts
Actuator / output	Performs a physical action if required

A real design must choose components and protocols based on requirements, power, accuracy, cost, and environment.

5.1 High-Level Architecture

Conceptual IoT data flow
PHYSICAL CONDITION
↓ sensed by
↓ read by
↓ optional network transfer
↓ sends data to
↓ optional response

This is a general teaching model. It does not indicate that the learner assembled, programmed, or deployed these components.

5.2 Low-Level Design Considerations

The circuit-design training highlighted that component selection and signal behavior must be considered before connecting a device to an IoT application.

• Electrical ratings: operate components within specified voltage and current ranges.

• Resistors and Ohm’s law: reason about voltage, current, and resistance.

• Signal levels and noise margin: preserve valid digital logic levels.

• Pull-up/pull-down resistors: avoid undefined digital input states where required.

• Filters and loading effect: consider signal variation and interaction between circuit stages.

• Fan-in/fan-out: consider the loads a logic output can reliably support.

• ADC/DAC: consider signal range and conversion resolution where applicable.

These are design considerations studied in training, not measured results from a specific circuit.

5.3 Interfaces, Protocols, and Data Flow

An IoT solution may combine device interfaces, communication links, and cloud APIs. Choices depend on range, data rate, power budget, reliability, and environment.

The supplied sample report mentions MQTT, CoAP, HTTP, Modbus TCP, and OPC UA in the context of industrial IoT platforms. These are examples from the template, not protocols used by a learner-built project.

Typical data flow: sensor reading → controller processing → optional filtering → network transfer → cloud storage or analytics → dashboard, notification, or command. Some systems operate locally without cloud connectivity for every control action.

Deployment planning should consider connectivity loss, data protection, and device maintenance.

6. Assessment and Testing

The learner reported completing IoT Test 2. No score, hardware test log, or project performance results were supplied. This section records the assessment activity and proposes ways to evaluate understanding rather than claiming measured device performance.

Knowledge can be checked through exercises such as identifying component roles, applying Ohm’s law, distinguishing sensors and actuators, explaining ADC/DAC, and tracing a conceptual IoT data path.

For a future hands-on project, testing should include component-level checks, signal checks, controller logic, connectivity where relevant, and end-to-end validation.

6.1 Test Plan / Knowledge Checks

ID	Check	Expected evidence	Status
K01	Voltage/current/resistance and Ohm's law	Explanation or worked exercise	Studied
K02	Identify electronic components	Describe component roles	Studied
K03	Sensor versus actuator	Correct input/output example	Studied
K04	Explain ADC and DAC	Describe conversion purpose	Studied
K05	IoT deployment challenges	Identify constraints and mitigations	Studied
K06	IoT Test 2	Assessment attempt as reported	Completed

"Studied" reflects the learner's description; it does not imply a separate practical hardware test.

6.2 Testing Procedure for a Future Prototype

• Write requirements and expected behavior before selecting hardware.

• Check component ratings and wiring against reliable documentation.

• Test the sensor or input circuit independently before integrating the controller.

• Check I/O behavior and conversion ranges under suitable test conditions.

• If networking is used, test normal operation and temporary disconnection.

• If a dashboard is used, compare displayed data with source readings.

• Record observed results, failures, corrections, and limitations.

No prototype test should be reported as completed until the system has been assembled or simulated and results recorded.

6.3 Assessment Outcome and Deployment Challenges

The reported assessment milestone is completion of IoT Test 2. A numerical score was not provided and is not included.

General deployment considerations include sensor suitability, gateway selection, connectivity, cloud platform, power requirements, environmental conditions, maintenance, and reliability.

A sensible engineering approach is to define requirements, compare options, prototype the smallest useful system, and test under realistic conditions. For this training-only internship, these remain recommendations rather than results from a deployed system.

7. My Learnings

The training strengthened my understanding that IoT combines electronics, embedded processing, communication, and application services. A physical measurement must be handled correctly before cloud analysis can be useful.

Key learnings: electrical concepts support circuit reasoning; sensors measure while actuators act; ADC/DAC explain analog and digital representation; noise margin, filters, loading, and pull resistors affect reliability; and IoT architecture requires decisions about devices, gateways, connectivity, cloud, and applications.

Deployment planning must consider practical constraints. Quizzes and case studies help connect terminology with real-world situations.

8. Future Work Scope

Since no individual IoT project was assigned, future work could apply the training to a small prototype. Possible learning projects aligned with the syllabus include weather monitoring, smart irrigation, air-quality monitoring, home intrusion detection, or a simulated IoT data pipeline.

These are suggestions, not completed projects. A future project should define the problem, required measurements, hardware, data flow, safety constraints, test cases, and success criteria.

Appendix: Repository and Submission Checklist

Student: Gayatri
Institution: UIET, CSJMU Kanpur
Internship: Embedded Systems and IoT
Status: Training and assignments only; no separate project assigned

Repository: https://github.com/gayatri-535/upskillcampus

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Expected report link after upload:
https://github.com/gayatri-535/upskillcampus/blob/main/EmbeddedIoT_Gayatri_USC_UCT.pdf

After uploading the PDF, open it on GitHub and verify the link. If a practical project is completed later, update the report with the actual problem statement, implementation details, diagrams, and real test results.